



Mars winds will clean Insight

NASA's Insight has gathered lots of dust, but local Martian winds might soon clean it up. <https://digit.in/june19-52>



New orbit for Earth?

This is an interesting read which debates whether the Earth could be moved of its orbit. <https://digit.in/june19-53>



Pillars of Light

This multicoloured optical atmospheric phenomena appears as vertical columns of light

Aditya Madanapalle | aditya@digit.in

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ight sources, thunderclouds and high energy particles interacting with the atmosphere cause all

kinds of optical phenomena, and even the most common of them, such as rays and rainbows are breathtaking. Some of the rarer ones include mirages, a green flash just after the sun has set, or the elusive aurora known as Steve. One of the rarer optical phenomena are called light pillars, which essentially are columns of light that seem to spring out of nowhere. Unlike beams of light, these pillars do not have a start or a definite end. High altitude clouds such as cirrus or cirrostratus clouds, have a lot of ice crystals in them. The smallest of these are commonly known as snowflakes.

These snowflakes have a hexagonal structure as a rule, which is caused by the way water deposits as a solid around a particle of dust, high in the atmosphere. The hexagonal snowflakes are the rudimentary forms of ice crystals, which can then grow into all kinds of symmetrical shapes, including columns, plates and needles. Light sources interacting with the hexagonal snowflakes still in clouds, cause a variety of atmospheric optical effects, including halos, which appear as giant rings around the light source, such as the Sun. These halos can appear as arcs, spots and even columns.

DIAMOND DUST

Light pillars appear under very specific circumstances. If the tempera-

tures are below the freezing point of water, which typically occurs in the highest and lowest latitudes, the ice crystals can settle at lower altitudes than is normal. Essentially, this results in a ground level cloud of ice crystals, which appears as a fog. This atmospheric phenomenon is known as diamond dust. The clouds of diamond dust can settle in one place for days at a time. If the diamond dust is made up of hexagonal or cylinder-shaped crystals, they can bounce back the light from artificial light sources to an observer at ground level. The reflections from the millions of ice crystals are only visible at very specific angles, and so always appears as vertical formations. These light pillars can be in various colours, depending on the colours of